

1351. Count Negative Numbers in a Sorted Matrix:

Given a `m x n` matrix `grid` which is sorted in non-increasing order both row-wise and column-wise, return *the number of **negative** numbers in `grid`*.

Example 1:

Input: `grid = [[4,3,2,-1],[3,2,1,-1],[1,1,-1,-2],[-1,-1,-2,-3]]`

Output: 8

Explanation: There are 8 negatives number in the matrix.

Example 2:

Input: `grid = [[3,2],[1,0]]`

Output: 0

Constraints:

- `m == grid.length`
- `n == grid[i].length`
- `1 <= m, n <= 100`
- `100 <= grid[i][j] <= 100`

Solution: Time complexity - $O(m*n)$

```
class Solution:
    def countNegatives(self, grid: List[List[int]]) -> int:
        count = 0
        for i in grid:
            for j in i:
                if j < 0:
```

```
        count += 1
    return count
```

time - complexity: $O(m+n)$ $\Rightarrow$ this is greedy algorithm.

```
class Solution:
    def countNegatives(self, grid: List[List[int]]) -> int:
        rows, cols = len(grid), len(grid[0])
        r, c = 0, cols - 1
        count = 0

        while r < rows and c >= 0:
            if grid[r][c] < 0:
                count += rows - r
                c -= 1
            else:
                r += 1

        return count
```